

The University of Jordan (UJ)
School of Engineering
Department of Computer Engineering
Advanced Networks Lab 0907529
Exp.9 Secure Remote Access and Virtual Private Network (VPN)

Objectives

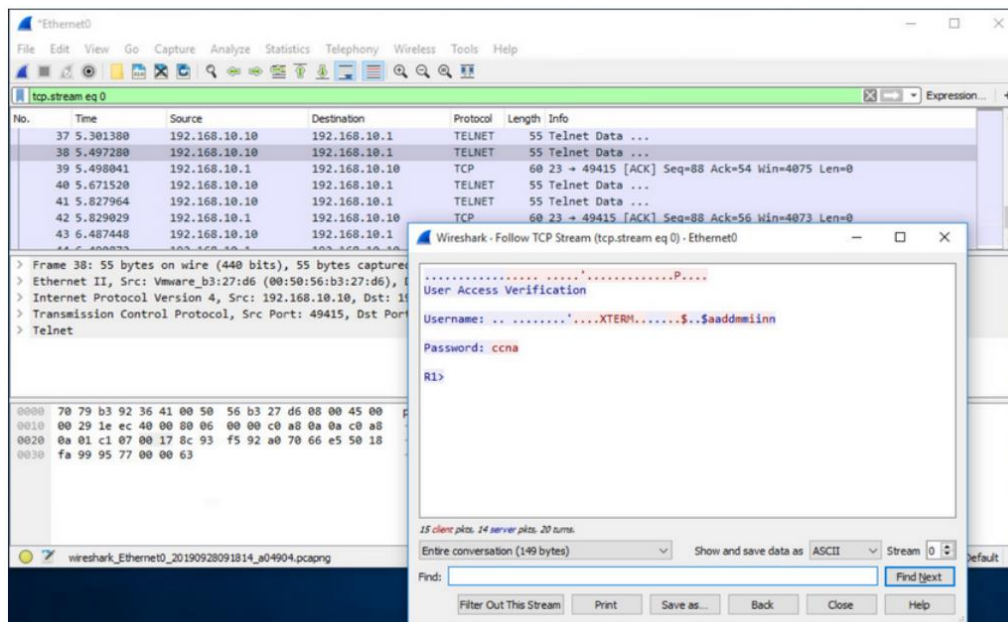
1. Discuss how to configure Secure Shell (SSH) for remote access.
2. Describe benefits of VPN technology.
3. Describe different types of VPNs.
4. Explain how the IPsec framework is used to secure network traffic.

Secure Remote Access using SSH

1. Introduction

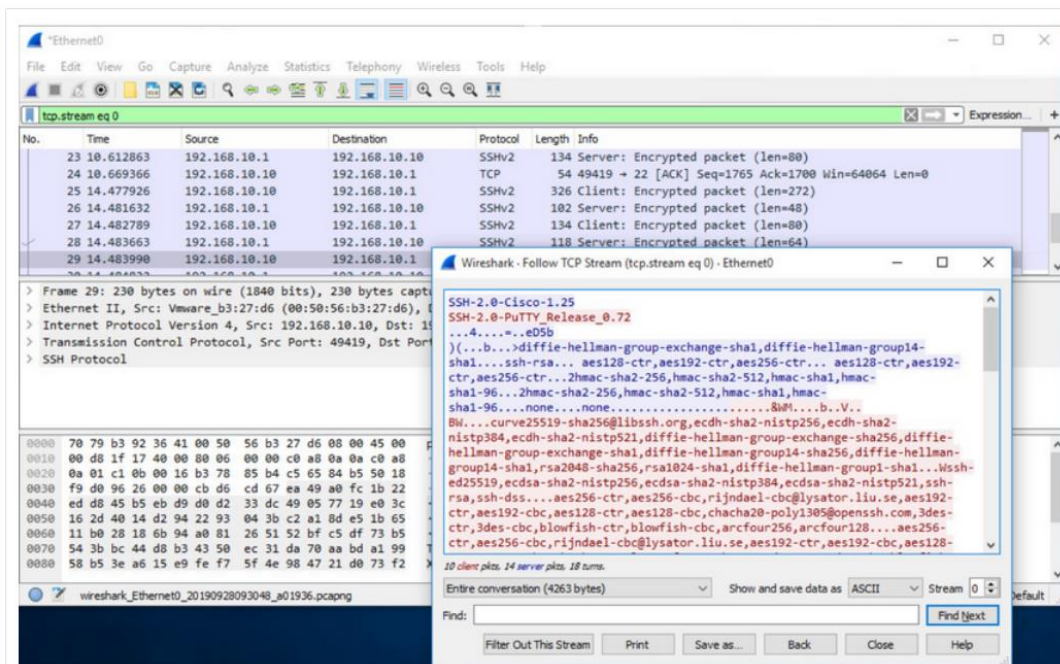
You might not always have direct access to your switch or router when you need to configure it. You need to be able to access it remotely and it is imperative that your access is secure. This topic discusses how to configure Secure Shell (SSH) for remote access.

Telnet uses TCP port 23. It is an older protocol that uses unsecure plaintext transmission of both the login authentication (username and password) and the data transmitted between the communicating devices. A threat actor can monitor packets using Wireshark. For example, in the figure the threat actor captured the username admin and password ccna from a Telnet session.



Secure Shell (SSH) is a secure protocol that uses TCP port 22. It provides a secure (encrypted) management connection to a remote device. SSH should replace Telnet for management connections. SSH provides security for remote connections by providing strong encryption when a device is authenticated (username and password) and also for the transmitted data between the communicating devices. For example, the figure shows a Wireshark capture of an SSH session.

The threat actor can track the session using the IP address of the administrator device. However, unlike Telnet, with SSH the username and password are encrypted.



2. SSH Configuration

Remote access tool
encrypted Text.

Before configuring SSH, the switch must be minimally configured with a unique hostname and the correct network connectivity settings.

Step 1: Verify SSH support.

Use the **show ip ssh** command to verify that the switch supports SSH. If the switch is not running an IOS that supports cryptographic features, this command is unrecognized. If the IOS filename includes the combination “k9” then it supports cryptographic (encrypted) features and capabilities.

```
S1# show ip ssh
```

Step 2: Configure the IP domain.

Configure the IP domain name of the network using the **ip domain-name domain-name** global configuration mode command. In the figure, the domain-name value is **cisco.com**.

```
S1(config)# ip domain-name cisco.com
```

Step 3: Generate RSA key pairs.

Not all versions of the IOS default to SSH version 2, and SSH version 1 has known security flaws. To configure SSH version 2, issue the **ip ssh version 2** global configuration mode command. Generating an RSA key pair automatically enables SSH. Use the **crypto key generate rsa** global configuration mode command to enable the SSH server on the switch and generate an RSA key pair. When generating RSA keys, the administrator is prompted to enter a modulus length. The sample configuration in the figure uses a modulus size of 1,024 bits. A longer modulus length is more secure, but it takes longer to generate and to use.

```
S1(config)# crypto key generate rsa
How many bits in the modulus [512]: 1024
```

Step 4: Configure user authentication.

The SSH server can authenticate users locally or using an authentication server. To use the local authentication method, create a username and password pair using the **username username secret password** global configuration mode command. In the example, the user admin is assigned the password ccna.

* S1(config)# username admin secret ccna

Step 5: Configure the vty lines.

Enable the SSH protocol on the vty lines by using the **transport input ssh** line configuration mode command. The Catalyst 2960 has vty lines ranging from 0 to 15. This configuration prevents non-SSH (such as Telnet) connections and limits the switch to accept only SSH connections. Use the **line vty** global configuration mode command and then the **login local** line configuration mode command to require local authentication for SSH connections from the local username database.

* S1(config)# line vty 0 15
S1(config-line)# transport input ssh No telnet!
S1(config-line)# login local
S1(config-line)# exit

S
0-16
R
0-4

Step 6: Enable SSH version 2.

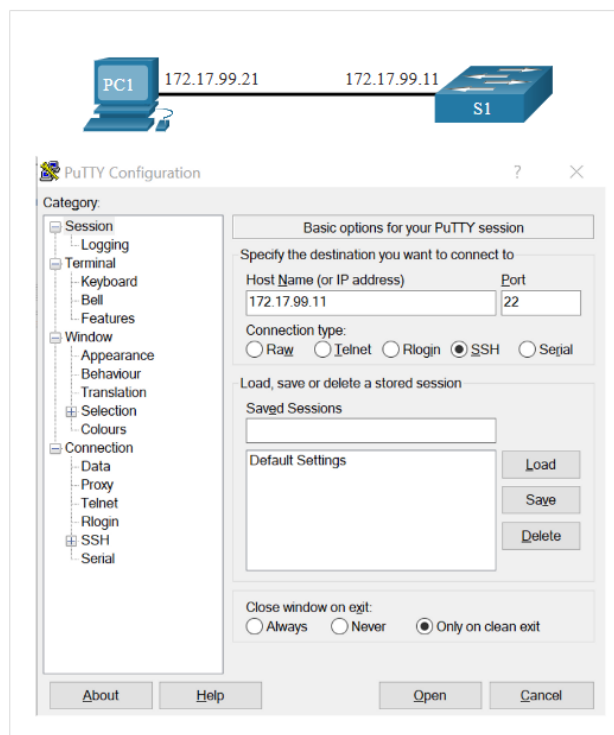
By default, SSH supports both versions 1 and 2. When supporting both versions, this is shown in the **show ip ssh** output as supporting version 2. Enable SSH version using the **ip ssh version 2** global configuration command.

* S1(config)# ip ssh version 2 if switch

3. Verify SSH is Operational

On a PC, an SSH client such as PuTTY, is used to connect to an SSH server. For example, assume the following is configured: SSH is enabled on switch S1, Interface VLAN 99 (SVI) with IPv4 address 172.17.99.11 on switch S1 and PC1 with IPv4 address 172.17.99.21

The figure shows the PuTTY settings for PC1 to initiate an SSH connection to the SVI VLAN IPv4 address of S1.



When connected, the user is prompted for a username and password as shown in the example. Using the configuration from the previous example, the username admin and password ccna are entered. After entering the correct combination, the user is connected via SSH to the command line interface (CLI) on the Catalyst 2960 switch.

```
Login as: admin
Using keyboard-interactive
Authentication.
Password:
S1> enable
Password:
S1#
```

To display the version and configuration data for SSH on the device that you configured as an SSH server, use the show ip ssh command. In the example, SSH version 2 is enabled.

```
S1# show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
To check the SSH connections to the device, use the show ssh command as shown.
S1# show ssh
%No SSHv1 server connections running.
Connection Version Mode Encryption Hmac State Username
0 2.0 IN aes256-cbc hmac-sha1 Session started admin
0 2.0 OUT aes256-cbc hmac-sha1 Session started admin
S1#
```

Virtual Private Network (VPN)

1. Introduction

Have you, or someone you know, ever been hacked while using public WiFi? It's surprisingly easy to do. But there is a solution to this problem: Virtual Private Networks (VPNs) and the additional protection of IP Security (IPsec). VPNs are commonly used by remote workers around the globe. There are also personal VPNs that you can use when you are on public WiFi. In fact, there are many different kinds of VPNs using IPsec to protect and authenticate IP packets between their source and destination.

2. VPN Benefits

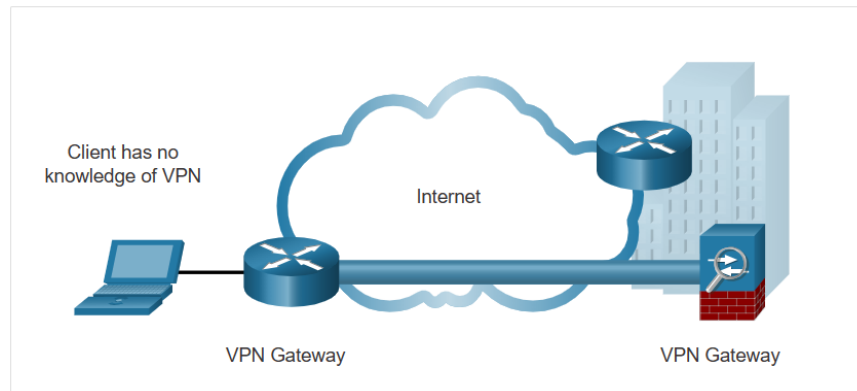
Major benefits of VPNs are shown in the table.

Benefit	Description
Cost Savings	With the advent of cost-effective, high-bandwidth technologies, organizations can use VPNs to reduce their connectivity costs while simultaneously increasing remote connection bandwidth.
Security	VPNs provide the highest level of security available, by using advanced encryption and authentication protocols that protect data from unauthorized access.
Scalability	VPNs allow organizations to use the internet, making it easy to add new users without adding significant infrastructure.
Compatibility	VPNs can be implemented across a wide variety of WAN link options including all the popular broadband technologies. Remote workers can take advantage of these high-speed connections to gain secure access to their corporate networks.

3. VPN Types

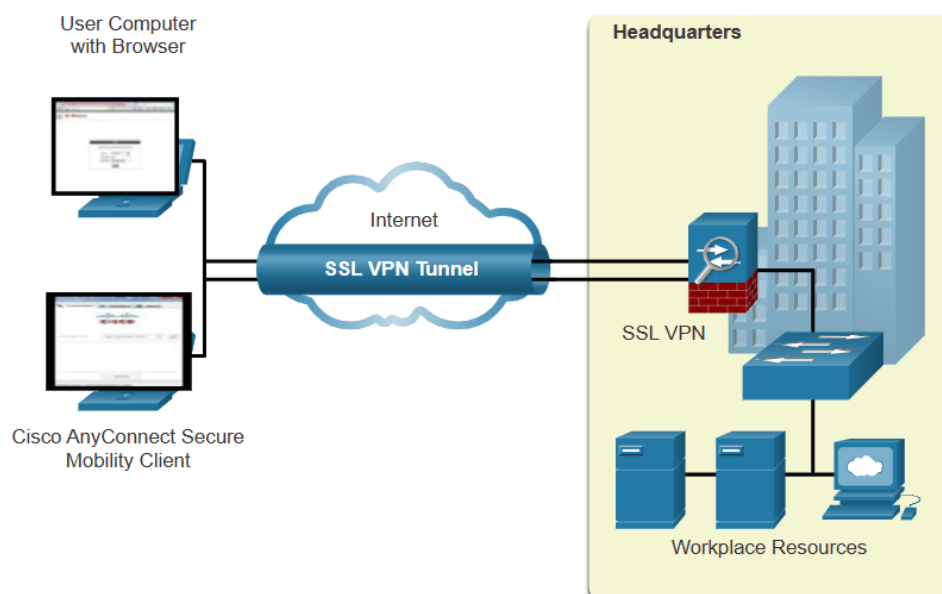
VPNs are commonly deployed in one of the following configurations: site-to-site or remote-access.

A site-to-site VPN is created when VPN terminating devices, also called VPN gateways, are preconfigured with information to establish a secure tunnel. VPN traffic is only encrypted between these devices. Internal hosts have no knowledge that a VPN is being used.



A remote-access VPN is dynamically created to establish a secure connection between a client and a VPN terminating device. As shown in the figure, remote-access VPNs let remote and mobile users securely connect to the enterprise by creating an encrypted tunnel. Remote users can securely replicate their enterprise security access including email and network applications. Remote-access VPNs also allow contractors and partners to have limited access to the specific servers, web pages, or files as required. This means that these users can contribute to business productivity without compromising network security.

Remote-access VPNs are typically enabled dynamically by the user when required. Remote access VPNs can be created using either IPsec or SSL. The figure displays two ways that a remote user can initiate a remote access VPN connection: clientless VPN (using web browser) and client-based VPN (using Cisco AnyConnect software).



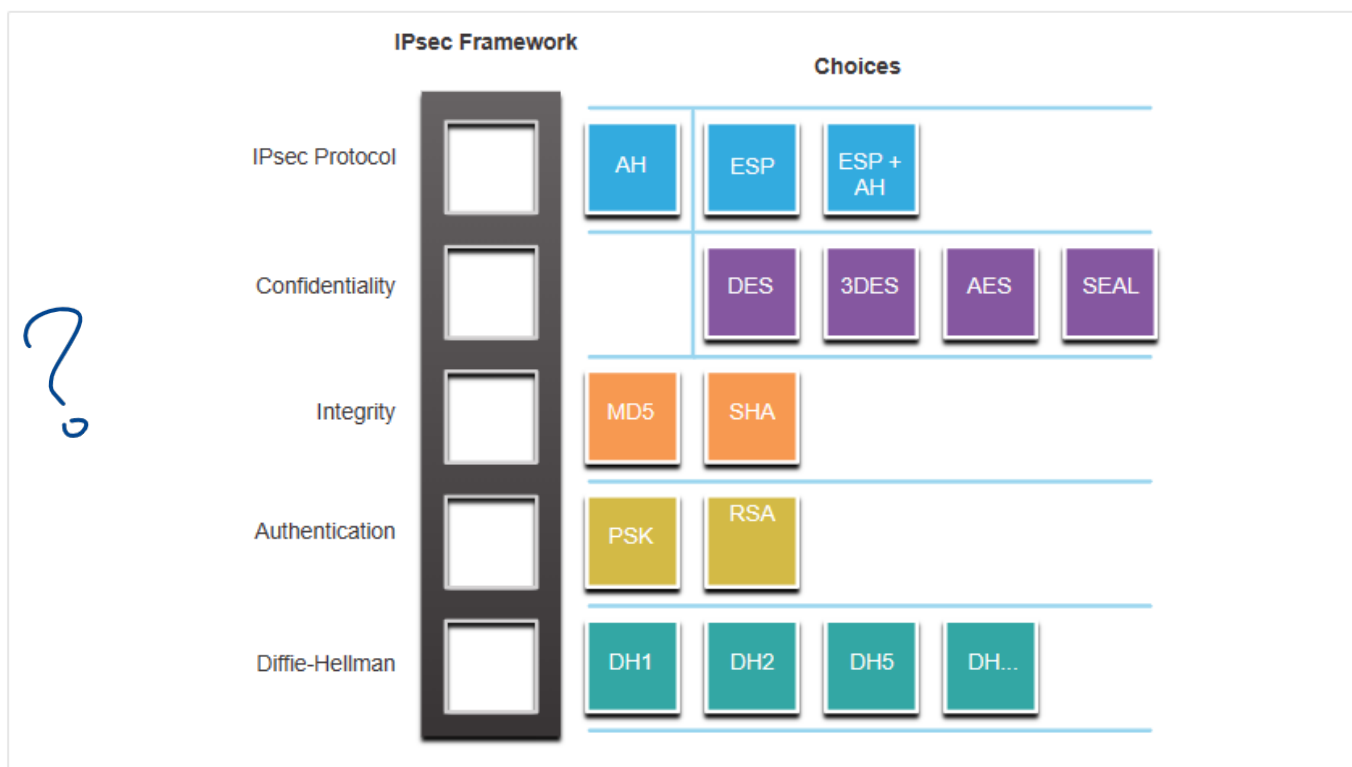
4. Site-to-Site IPsec VPNs

Site-to-site VPNs are used to connect networks across another untrusted network such as the internet. The VPN gateway encapsulates and encrypts outbound traffic. It then sends the traffic through a VPN tunnel over the internet to a VPN gateway at the target site. Upon receipt, the receiving VPN gateway strips the headers, decrypts the content, and relays the packet toward the target host inside its private network. Site-to-site VPNs are typically created and secured using IP security (IPsec).

Other VPN tunneling protocols are Generic Routing Encapsulation (GRE), Dynamic Multipoint VPN (DMVPN) and IPsec Virtual Tunnel Interface (VTI). We will not discuss these types in this experiment.

IPsec protects and authenticates IP packets between source and destination. IPsec can protect traffic from Layer 4 through Layer 7. Using the IPsec framework, IPsec provides these essential security functions:

- **Confidentiality** - IPsec uses encryption algorithms to prevent cybercriminals from reading the packet contents.
- **Integrity** - IPsec uses hashing algorithms to ensure that packets have not been altered between source and destination.
- **Origin authentication** - IPsec uses the Internet Key Exchange (IKE) protocol to authenticate source and destination. Methods of authentication including using pre-shared keys (passwords), digital certificates, or RSA certificates.
- **Diffie-Hellman** - Secure key exchange typically using various groups of the DH algorithm.



IPsec is not bound to any specific rules for secure communications. This flexibility of the framework allows IPsec to easily integrate new security technologies without updating the existing IPsec standards. The currently available technologies are aligned to their specific security function. The open slots shown in the IPsec framework in the figure can be filled with any of the choices that are available for that IPsec function to create a unique security association (SA).

The security functions are listed in the table.

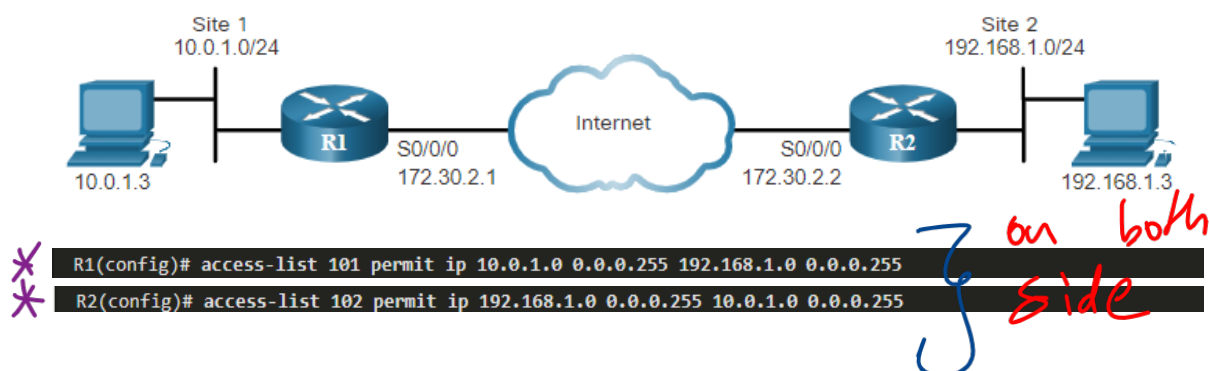
IPsec Function	Description
IPsec Protocol	The choices for IPsec Protocol include Authentication Header (AH) or Encapsulation Security Protocol (ESP). AH authenticates the Layer 3 packet but doesn't use any encryption method. ESP encrypts the Layer 3 packet. Note: ESP+AH is rarely used as this combination will not successfully traverse a NAT device.
Confidentiality	Encryption ensures confidentiality of the Layer 3 packet. Choices include Data Encryption Standard (DES), Triple DES (3DES), Advanced Encryption Standard (AES), or Software-Optimized Encryption Algorithm (SEAL). No encryption is also an option.
Integrity	Ensures that data arrives unchanged at the destination using a hash algorithm, such as message-digest 5 (MD5) or Secure Hash Algorithm (SHA).
Authentication	IPsec uses Internet Key Exchange (IKE) to authenticate users and devices that can carry out communication independently. IKE uses several types of authentication, including username and password, one-time password, biometrics, pre-shared keys (PSKs), and digital certificates using the Rivest, Shamir, and Adleman (RSA) algorithm.
Diffie-Hellman	IPsec uses the DH algorithm to provide a public key exchange method for two peers to establish a shared secret key. There are several different groups to choose from including DH14, 15, 16 and DH 19, 20, 21 and 24. DH1, 2 and 5 are no longer recommended.

5. IPsec Configuration

In this part, we will show you how to configure two Cisco IOS routers to use IPsec in Tunnel mode.

Step 1: ACL

To define interesting traffic, configure each router with an ACL to permit traffic from the local LAN to the remote LAN, as shown in the following examples for R1 and R2. The ACL will be used in the crypto map configuration to specify what traffic will trigger the start of IKE Phase 1.



Step 2: Configure ISAKMP Policy (Phase 1)

between Routers !.

To configure a new ISAKMP policy, use the **crypto isakmp policy** command, as shown in the figure. The only argument for the command is to set a priority for the policy (from 1 to 10000). Peers will attempt to negotiate using the policy with the lowest number (highest priority). Peers do not require matching priority numbers.

When in ISAKMP policy configuration mode, the SAs for the IKE Phase 1 tunnel can be configured. Use the mnemonic **HAGLE** to remember the five SAs to configure:

Hash
Authentication
Group
Lifetime
Encryption

The example below shows the ISAKMP policy configuration. Use the **show crypto isakmp policy** command to verify the configuration. R2 has an equivalent configuration.

```
R1(config)# crypto isakmp policy 1
R1(config-isakmp)# encryption aes 256
R1(config-isakmp)# hash sha
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# group 24
R1(config-isakmp)# lifetime 3600
R1(config-isakmp)# end
R1# show crypto isakmp policy

Global IKE policy
Protection suite of priority 1
  encryption algorithm: AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #24 (2048 bit, 256 bit subgroup)
  lifetime:            3600 seconds, no volume limit

R1#
```

```
R2(config)# crypto isakmp policy 1
R2(config-isakmp)# hash sha
R2(config-isakmp)# authentication pre-share
R2(config-isakmp)# group 24
R2(config-isakmp)# lifetime 3600
R2(config-isakmp)# encryption aes 256
```

Security policy requires that a pre-shared key be used for authentication between the peers. The administrator can either specify a host name or an IP address for the peer. The command syntax is shown below.

```
Router(config)# crypto isakmp key keystring address peer-address
Router(config)# crypto isakmp key keystring hostname peer-hostname
```

```
* R1# conf t
R1(config)# crypto isakmp key cisco12345 address 172.30.2.2
R1(config)#
```

الاول

```
R2# conf t
R2(config)# crypto isakmp key cisco12345 address 172.30.2.1
R2(config)#
```


Step 3: Configure the Transform Set (Phase 2) *from private Networks.*

The next step is to configure the set of encryption and hashing algorithms that will be used to transform the data sent through the IPsec tunnel. This is called the transform set. During IKE Phase 2 negotiations, the peers agree on the IPsec transform set to be used for protecting interesting traffic.

Configure a transform set using the **crypto ipsec transform-set** command, as shown here. First, specify a name for the transform set. After the transform set is named, the encryption and hashing algorithm can be configured in either order.



```
R1(config)# crypto ipsec transform-set R1-R2 esp-aes esp-sha-hmac
R1(config)#
```

```
R2(config)# crypto ipsec transform-set R1-R2 esp-aes esp-sha-hmac
R2(config)#
```

Step 4: Configure IPsec Crypto Map (tie it together)

Now that the interesting traffic is defined, and an IPsec transform set is configured, it is time to bind those configurations with the rest of the IPsec policy in a crypto map. The syntax to start a crypto map set is shown below. The sequence number is important when configuring multiple crypto map entries.

To finish the configuration to meet the IPsec security policy, complete the following:

- Step 1. Bind the ACL and the transform set to the map.
- Step 2. Specify the peer's IP address.
- Step 3. Configure the DH group.
- Step 4. Configure the IPsec tunnel lifetime.

The crypto map configurations for R1 and R2 are shown below. The map name is **R1-R2_MAP**, and the sequence number is **10**.

p2 p1

```
* R1(config)# crypto map R1-R2_MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
* R1(config-crypto-map)# match address 101 access control list
* R1(config-crypto-map)# set transform-set R1-R2
* R1(config-crypto-map)# set peer 172.30.2.2
* R1(config-crypto-map)# set pfs group24
* R1(config-crypto-map)# set security-association lifetime seconds 900
R1(config-crypto-map)# exit
R1(config)#
```

```
R2(config)# crypto map R1-R2_MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R2(config-crypto-map)# match address 102
R2(config-crypto-map)# set transform-set R1-R2
R2(config-crypto-map)# set peer 172.30.2.1
R2(config-crypto-map)# set pfs group24
R2(config-crypto-map)# set security-association lifetime seconds 900
R2(config-crypto-map)# exit
R2(config)#
```

Use the **show crypto map** command to verify the crypto map configuration, as shown below for R1. All the required SAs should be in place. Notice that the output shows that no interfaces are currently using the crypto map.

```
R1# show crypto map
Crypto Map IPv4 "R1-R2_MAP" 10 ipsec-isakmp
  Peer = 172.30.2.2
  Extended IP access list 101
    access-list 101 permit ip 10.0.1.0 0.0.0.255 192.168.1.0 0.0.0.255
  Security association lifetime: 4608000 kilobytes/900 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): Y
  DH group: group24
  Mixed-mode : Disabled
  Transform sets={
    R1-R2: { esp-aes esp-sha-hmac } ,
  }
  Interfaces using crypto map R1-R2_MAP:

R1#
```

Step 5: Applying Crypto Maps to Interfaces

To apply the crypto map, enter interface configuration mode for the outbound interface and configure the **crypto map map-name** command. Below is the configuration. Notice the **show crypto map** output now displays that the Serial 0/0/0 interface is using the crypto map. R2 is configured with the same command on its Serial 0/0/0 interface.

```
R1(config)# interface serial0/0/0
R1(config-if)# crypto map R1-R2_MAP
R1(config-if)#
*Mar 19 19:36:36.273: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
R1(config-if)# end
R1# show crypto map
Crypto Map IPv4 "R1-R2_MAP" 10 ipsec-isakmp
  Peer = 172.30.2.2
  Extended IP access list 101
    access-list 101 permit ip 10.0.1.0 0.0.0.255 192.168.1.0 0.0.0.255
  Security association lifetime: 4608000 kilobytes/900 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): Y
  DH group: group24
  Mixed-mode : Disabled
  Transform sets={
    R1-R2: { esp-aes esp-sha-hmac } ,
  }
  Interfaces using crypto map R1-R2_MAP:
    Serial0/0/0
```

Now that both the ISAKMP and IPsec policies are configured, and the crypto map is applied to the appropriate outbound interfaces, test the two tunnels by sending interesting traffic across the link.

Traffic from the LAN interface on R1 that is destined for the LAN interface on R2 is considered interesting traffic because it matches the ACLs configured on both routers. An extended ping from R1 will effectively test the VPN configuration. The extended ping command syntax and results are shown below. The first ping failed because it takes a few milliseconds to establish the ISAKMP and IPsec tunnels.

```

R1# ping 192.168.1.1 source 10.0.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
Packet sent with a source address of 10.0.1.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/4 ms
R1#

```

Sending interesting traffic does not actually mean that the tunnels are established. R1 and R2 will route traffic between the two LANs even if the ISAKMP and IPsec policy configurations are wrong. To verify that tunnels have been established, use the **show crypto isakmp sa** and **show crypto ipsec sa** commands. In the output below, notice that the tunnel is active between the two peers, 172.30.2.1 and 172.30.2.2, and that they are using the R1-R2_MAP crypto map.

```

R1# show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state      conn-id status
172.30.2.2   172.30.2.1   QM_IDLE    1005 ACTIVE

R1# show crypto ipsec sa

interface: Serial0/0/0
  Crypto map tag: R1-R2_MAP, local addr 172.30.2.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (10.0.1.0/255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.1.0/255.255.0/0/0)
current_peer 172.30.2.2 port 500

```

Procedures:

Dear students, please note that the lab problems sheet, the packet tracer activities and the practical discussion videos have been uploaded on your Microsoft Teams group. You are required to carefully study this experiment and then complete the lab sheet.

References

Network Security - Cisco Networking Academy
<https://www.netacad.com>